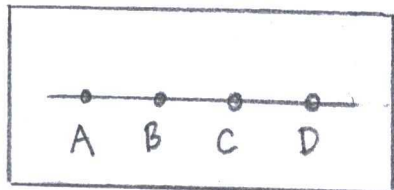


Euclidean and NonEuclidean Geometry:

Exercises on Betweenness:

1.



a) A, B, C, and D are four distinct points:

Proof: Hypothesis	Evidence
1) From points B and D, other points reside on $\overleftrightarrow{BD}$, such as A and C.	Axiom B-2

b) A, B, C, and D are collinear:

Proof: Hypothesis	Evidence
1) All points on line $\overleftrightarrow{BD}$ belong to segment BD on $\overleftrightarrow{BD}$	Euclid's Postulate <u>1</u>

c) (pg 110) "Corollary to Axiom B-4"

"For every line ℓ and for any three points A, B, and C not lying on ℓ

(i) If A and B are on the same side of ℓ and if B and

C are on the same side of l , then A and C are on the same side of l

(ii) If A and B are on opposite sides of l and if B and C are on opposite sides of l , then A and C are on the same side of l .

(iii) If A and B are on same side of l and B and C are on opposite sides of l , then A and C are on opposite sides of l . "

Proof:	Hypothesis	Evidence
1)	If B and C were on opposite sides of l , then A and C are on the same side	Axiom B-4(ii)
2)	A and C are on opposite sides of l	False RAA

2.

a) (pg 109) "Proposition 3.1"

"For any two points A and B

$$(i) \overrightarrow{AB} \cap \overrightarrow{BA} = AB$$

$$(ii) \overrightarrow{AB} \cup \overrightarrow{BA} = \{\overrightarrow{AB}\}$$

Proof :	Hypothesis	Evidence
1)	$AB \subset \overrightarrow{AB}$ and $AB \subset \overrightarrow{BA}$	(pg 16) "Definition - Segment"
2)	$AB \subset \overrightarrow{AB} \cap \overrightarrow{BA}$	(pg 18) "Definition - Ray"
3)	Point C belongs to segment AB	(pg 45) "Definition - Intersection"
4)	A, B, and C are collinear points, and $A \times C \times B, A \times B \times C$ or $C \times A \times B$	RAA Premise
5)	i) Point C is on $\overrightarrow{AB}$ for $A \times B \times C$	Axiom B-1
		Axiom B-3
		(pg 16) "Definition - Ray"

i) Point C is on $\overrightarrow{BA}$ for $B \ast A \ast C$

(pg 16) "Definition - Ray"

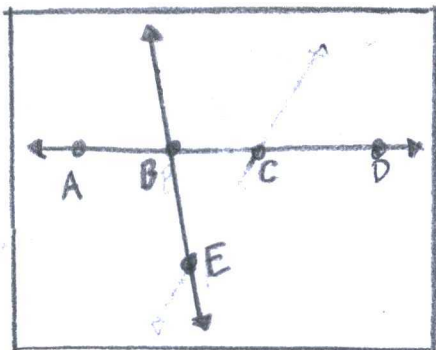
ii) Point C is between A and B for $A \ast C \ast B$

(pg 16) "Definition - Ray"

6) $A \ast B \ast C$, $C \ast A \ast B$ and $A \ast C \ast B$ are lines

Axiom B-1 (pg 12) "Definition - Lines"

b) (pg 113) "Proposition 3.3"



" $A \ast B \ast D$ "

"Given $A \ast B \ast C$ and $A \ast C \ast D$. Then $B \ast C \ast D$ and $A \ast B \ast D$ "

Proof	Hypothesis	Evidence
	1) A, B, C, and D are four distinct collinear points	
	2) Point E exists not on the line through A, B, C, D	Proposition 2.3

3) A line $\overleftrightarrow{EB}$ meets AD at point B. Also, A and D are on opposite sides of $\overleftrightarrow{EB}$

Contrary to RAA hypothesis

4) C and D are on opposite sides

5) Then $\overleftrightarrow{EB}$ meets CD in a point between C and D

(pg 110) "Definition-
opposite sides"

6) At point B

Proposition 2.1

7) Thus, $C * B * D$
and $B * C * D$

RAA falsified
because Axiom B-4

8) So, C and D are on the same side

RAA conclusion

9) A and C are on opposite sides to $\overleftrightarrow{EB}$

Steps #3-8
and corollary
to Axiom B-4

10) Point B is
the intersection
for the lines
 $\overleftrightarrow{EB}$ and $\overleftrightarrow{AD}$.
Between A and
C is point B.

(pg 110) "Definition -
opposite sides"
Proposition 2.1

c) Converse to Proposition 3.3.

Proof: Hypothesis

Evidence

1) Point C is the
intersection of
 $\overleftrightarrow{EC}$ and $\overleftrightarrow{BD}$. Point
C is between B
and D.

Proposition 2.1
(pg 110) "Definition -
opposite
sides"

2) B and D are on
opposite sides of
 $\overleftrightarrow{EC}$

Hypothesis

3) Points A and B
are on the same
side of $\overleftrightarrow{EC}$

Hypothesis

10) A, B, C, and D
are four distinct
collinear points

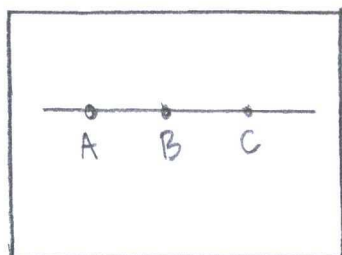
d) (pg 113) "Corollary to Proposition 3.3"

"Given $A \times B \times C$ and $B \times C \times D$. Then
 $A \times B \times D$ and $A \times C \times D$ "

Proof:	Hypothesis	Evidence
	1) Points B and C form a ray	(pg 12) "Definition-Line"
	2) Both points A and D are on $\overrightarrow{BC}$ or $\overrightarrow{CB}$	Proposition 3.1
	3) $A \times B \times D$ and $A \times C \times D$	Axiom B-1 and Axiom B-4

3.

a) $ABC \subseteq AC$



Proof:	Hypothesis	Evidence
	1) A, B, and C are collinear points	Axiom B-1

4) $A \times B \times C$ and $A \times C \times B$

False RAA

5) The point for
intersection is

C

6) $\overleftrightarrow{EC}$ meets $\overleftrightarrow{AB}$
in a point
between A and B

(pg 110) "Definition -
opposite
sides"

7) A and B must
be on opposite
sides the line
 $\overleftrightarrow{EC}$

False RAA

8) Points A and B
are on the same
side

True conclusion
from hypothesis,
so $A \times B \times C$

9) A and D are
on opposite sides
line $\overleftrightarrow{EC}$

Steps #3-8

10) Point E coincides
not on line through
A, B, C, D

Proposition 2.3

2) B is between
A and C

Euclid's
Postulate #2

b) $AC \subset AB \cup BC$

Proof: Hypothesis

Evidence

1) A, B, and C
are three collinear
points

Axiom B-1

2) $AB \cup BC$

a) $AB \subset \overrightarrow{AB}$ and
 $BC \subset \overrightarrow{BC}$

(pg 16) "Definition-
Rays"

(pg 16) "Definition-
Segments"

b) $\overrightarrow{AB} \cup \overrightarrow{BC}$

(pg 45) "Definition-
Union"

3) $\overrightarrow{AB} \cup \overrightarrow{BC} = \overrightarrow{AC}$

(pg 45) "Definition-
Union"

4) $\overrightarrow{AC} = \overrightarrow{AC}$

(pg 16) "Definition-
Rays"

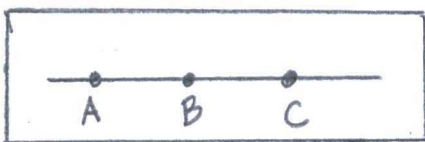
4) $\overrightarrow{AC} \subset AC$

c) (pg 114) "Proposition 3.5"

"Given $A * B * C$. Then $AC = AB \cup BC$
and B is the only point common
to segments AB and BC"

Proof: Hypothesis	Evidence
1) A point P exists on $A*B*C$.	Axiom B-1
2) Point P is not B	RAA conjecture
3) So, point P is on segment AB or BC	
4) Two lines are possible: $A*P*B$ and $B*P*C$	Axiom B-1
5) $AC = AB \cup BC$	
a) $A*P*B \subset A*B$	
b) $B*P*C \subset B*C$	
c) $AB \subset \overrightarrow{AB}$	"Definition - Ray"
d) $BC \subset \overrightarrow{BC}$	"Definition - Ray"
e) $AB \cup BC \subset \overrightarrow{AB \cup BC}$	"Definition - Union"
f) $\overrightarrow{AC} = \overrightarrow{AB \cup BC}$	
g) $\overrightarrow{AC} \subset AC$	"Definition -"
6) $P=B$ is	RAA contradiction

4.



$$a) \sim A*B*P \Rightarrow \sim A*C*P$$

Proof: Hypothesis

Evidence

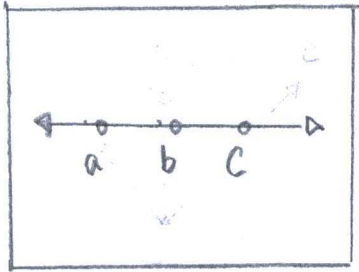
13,

A line cannot be contained in the interior of a triangle^s

Proof	Hypothesis	Evidence
	1) A, B, and C are distinct noncollinear points	
	2) Segments form between A, B, and C into AB, AC, and BC	(pg 16) "Definition-Segment"
	3) A point appears inside the triangle. The point label - P.	
	4) A line PQ is from point P to point Q.	(pg 12) "Definition-line"
	5) The line PQ intersects	Pasch's Theorem
	a) Point A	
	b) Point B	
	c) Point C	
	d) Segment AB	
	e) Segment AC	

f) Segment BC

14.



"coterminal"

a) Analogue to Axiom B-1:

"If $a*b*c$, then a, b, c are distinct and coterminal and $c*b*a$ "

Analogue to Axiom B-2:

"If rays b and d are distinct, then rays a, c , and e reside on $\overleftrightarrow{bd}$ as distinct rays, coterminal, also $a*b*d, b*c*d$, and $b*d*e$."

Analogue to Axiom B-3:

"If a, b , and c are on the same line, then one and only one ray is between the other two."

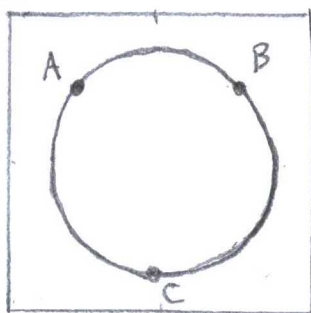
Analogue to Proposition 3.3:

"If $a*b*c$ and $a*c*d$, then $b*c*d$ and $a*b*d$."

The incorrect analogue parts are coterminal. Rays are between each other

and not coterminal.

15.



"three collinear points, no one of which is between the other two"

Betweenness. Axiom 3 (pg 108) fails interpretation in unclear circumstances about lines. Definitions consider both circular and curved lines. So, in Axiom 3, "no one of which is between the other two" is necessary.

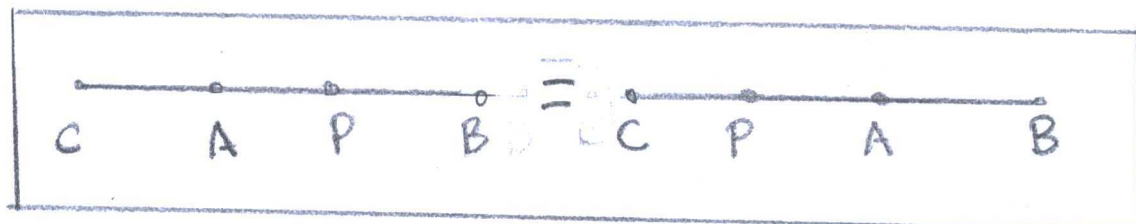
16.

Marvin Greenburg asked an "interpretation" to between. What about a line, such as $C \times A \times B$, with "between" defining a segment from point to point. The line $C \times A \times B$ is from the exercise. Axioms B-1, 2, 3 hold, but not Proposition 3.4 from the new definition - "between". An example, point P is between A and B, in addition, the point A is between both P and B. The segment $\overline{AP}$

is irregardless to endpoints.

Between A and P means

between P and A in layout,



17.

$$\frac{a}{2^n}$$

"Dyadic"

(pg 114) "Pasch's Theorem"

"IF A, B, C are distinct noncollinear points and l is any line

intersecting AB in a point

between A and B, then l also

intersects either AC and BC.

IF C does not lie on l , then

l does not intersect both AC

and BC."

The book presented an example

on dyadic points. Vertices are

at $(0,0)$, $(0,1)$, and $(\frac{1}{3},0)$. A line $3x+y=1$

supposedly never does

Scale in a dyadic plane.

If x -values scale by $n=0,1,2,3,\dots$

then the vertices too. The line.

models Pasch's theorem at $n=0$,

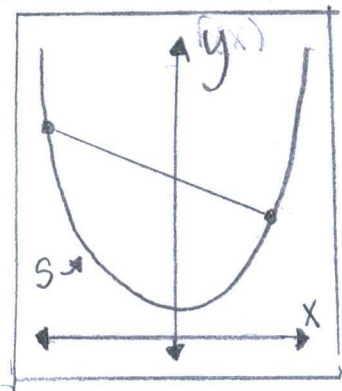
but not $n \geq 2$. Why? The line

becomes piecewise. The dyadic

function is $y = -\frac{3x}{2^n} + 1$ and

requires a decent amount of
 n -values.

10.



(from exercise) "Convex"

"Whenever two points A and B are
in S , the entire segment AB is
contained in S "

Half-plane is convex:

Proof:	Hypothesis	Evidence
	1) A half-plane has a line l	Proposition 3.2
	2) Set S for a half-plane is	$S = \{(x,y) \in \mathbb{R}^2\}$

the values greater
or less than line
 l

3) If points A and B
are greater than
 (x, y) , then also
segment AB

$$S = \{(x, y) \in \mathbb{R}^2 : ax + by + c \geq 0\}$$

4) If points A and
B are less than
 (x, y) , then also
segment AB

$$S = \{(x, y) \in \mathbb{R}^2 : ax + by + c \leq 0\}$$

Interior of an angle is convex:

Proof: Hypothesis

Evidence

1) The set S
of interior of
an angle is
less than 180°
and greater
than 0° .

$$S = \{x \in \mathbb{R} : 0 < x < 180\}$$

2. Set S contains
segment AB

When points A and B are greater than 0° and less than 180° . By notation, point A is greater in degree to B.

$$AB = \{A + tB \mid t \in [0, 1], A > B\} \in S$$

Interiors of a triangle are convex:

Proof: Hypothesis

Evidence

1) The set S of interior angles is less than 180° and greater than 0° .

$$S = \{x \in \mathbb{R} : 0 < x < 180\}$$

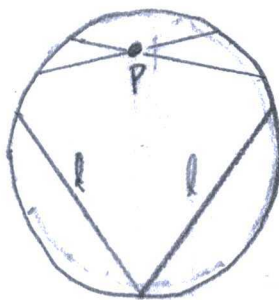
2) Set S contains segment AB when points A and B less than 180° and greater than 0° . For notation $A > B$.

$$AB = \{A + tB \mid t \in [0, 1], A > B\} \in S$$

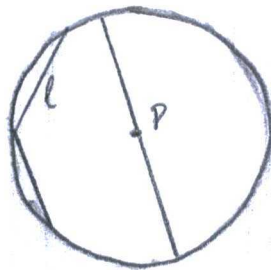
(pg 75) "Hyperbolic Parallel Property"

"For every line l and every point P not on l , there exist at least two lines through P parallel to l "

Examples showing Hyperbolic Parallel Property in Chords around a Unit Circle:



Author hearsays, "lines through point P are parallel to lines l ."



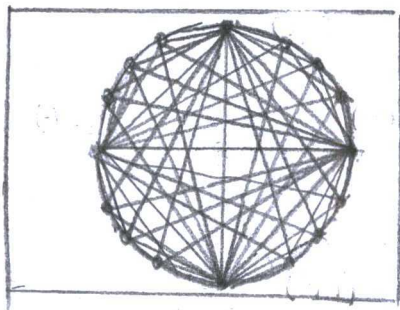
Supposedly, the chord through P is parallel to lines l .

Exterior points of a triangle
are not convex.

Proof: Hypothesis	Evidence
1) Exterior, set S of a triangle is all points outside a triangle from 0 to infinity.	$S = \{(x, y) \in \mathbb{R}^2 : x = (-\cos(1-u-v))x_1 + ux_2 + vx_3, y = (-\cos(1-u-v))y_1 + uy_2 + vy_3, u, v \geq 0, u+v \leq 1\}$ where $(x_1, y_1), (x_2, y_2)$ and (x_3, y_3) are vertices to triangle
2) If points A and B are outside the triangle, then not every segment AB.	$AB = \{A + t(B-A) t \in [0, 1], A > B\} \notin S.$

A triangle has finite points inside,
so convex.

19.



"Chords on a
Circle"

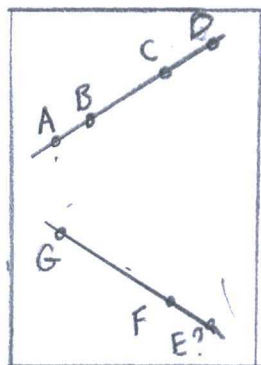
Exercises on Congruence:

20. (Pg 124) "Proposition 3.11 (Segment Subtraction)"

Proof:	Hypothesis	Evidence
1) Assume on the contrary that BC is not congruent to EF		RAA Hypothesis
2) Then there is a point G on EF such that $BC \cong EG$		Euclid Postulate II
3) $G \neq F$		(Pg 12) "Definition-point"
4) Since $AB \cong DE$, adding gives $AC \cong DG$		Axiom C-3
5) However, $AC \cong DF$		Contradiction
6) Hence, $DF \cong DG$		Requirement of Reality
7) Therefore, $F = G$		
8) Our assumption led to a contradiction; hence, $BC \cong EF$		False RAA result

21.

(Pg 125) "Proposition 3.13a (Segment Ordering)"



"Exactly one of the following statements holds (trichotomy):

$AB < CD$, $AB \cong CD$ or $AB > CD$."

Proof

Hypothesis

Evidence

1) $AB \cong CD$	RAA Hypothesis #1
a) A point E resides on AB and F on CD, such $AE \cong CF$	Proposition 3.12
b) $EB \cong FD$	Proposition 3.11
2) $AB < CD$	RAA Hypothesis #2
a) A point G exists on CD, such $AB \cong CG$	Euclid Postulate II
b) $D \neq G$	(pg 12) "Definition-Point"
3) $AB > CD$	RAA Hypothesis #3
a) A point H exists on AB, such $AH \cong CD$	Euclid Postulate II
b) $B \neq H$	
4) IF $D = G$ or $B = H$, then initial step #1.	False result

(pg 132) "Proposition 3.12"

"Given $AC \cong DF$, then for any point B between A and C , there is a unique point E between D and F such that $AB \cong DE$."

(pg 133) "Proposition 3.13b"

"If $AB < CD$ and $CD \cong EF$, then $AB < EF$ "

Proof:	Hypothesis	Evidence
	1) A point G appears on segment CD , such that $AB \cong EG$	Proposition 3.12
	2) A point H appears on EF , such that $AB \cong EH$	Proposition 3.12
	3) $GD \cong HF$	Proposition 3.11

(pg 133) "Proposition 3.13c"

"If $AB > CD$ and $CD \cong EF$, then $AB > EF$ "

Proof:	Hypothesis	Evidence
	1) A point G appears	Proposition 3.12

on segment

AB, such $AG \cong CD$

2) $AG \cong EF$ pairs

Axiom C-2

23.

(pg 113) "Proposition 3.13d"

"If $AB < CD$ and $CD < EF$, then $AB < EF$."

Proof: Hypothesis

Evidence

1) A point G resides on segment EF, such $CD \cong EG$

Proposition 3.12

2) Another point H resides on CD, such $AB \cong CH$

Proposition 3.12

3) $0 < HD < GF$

(pg 124) "Definition-Transitive Inequality"

24.

(pg 125) "Proposition 3.14"

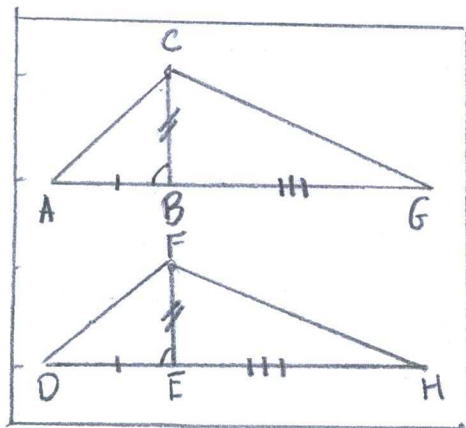
"Supplements of congruent angles are congruent."

Proof: Hypothesis

Evidence

1) Triangle $\triangle ABC \cong \triangle DEF$

Axiom C-6



"Same shade."

Where $\angle B = \angle E$	
2) A unique ray $\overrightarrow{AB}$ is on side AB	Axiom C-4
3) A unique ray $\overrightarrow{DE}$ is on side DE	Axiom C-4
4) A point G is on AB and a point H is on DE	Axiom B-2
5) H is on DE for $BG \cong EH$	Proposition 3.11
6) $AG \cong DH$	Axiom C-3
7) $\triangle ACG \cong \triangle DFH$	Axiom C-6 (SAS)
8) $\triangle CBG \cong \triangle FEH$	Axiom C-6 (SAS)
9) $\angle CBG \cong \angle FEH$	Pons Asinorum

25.

(pg 125) "Proposition 3.15"

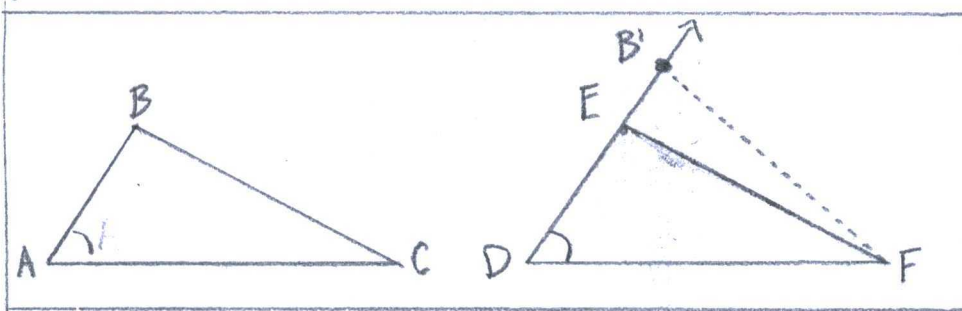
a) Vertical angles are congruent
to each other

b) An angle congruent to a right

angle are congruent to each other."

Euclid deduced in the fourth (IV) postulate congruent right angles. Specially, Proposition 3.14 justified right angles by the supplementary counterpart. -

26.



"ASA Criterion for Congruence"

(pg 127) "Proposition 3.17"

"Given $\triangle ABC$ and $\triangle DEF$ with $\angle A \cong \angle D$, $\angle C \cong \angle F$, and $AC \cong DF$. Then $\triangle ABC \cong \triangle DEF$."

Proof:	Hypothesis	Evidence
	1) There is a unique point B' on ray $\overrightarrow{DE}$ such that $DB' \cong AB$	Axiom C-1

$$2) \triangle ABC \cong \triangle DB'F$$

$$DB' \cong AB, \angle A \cong \angle D$$

$$AC \cong DF \text{ is}$$

Axiom C-6

$$3) \angle DFB' \cong \angle C$$

Axiom C-5

$$4) \text{ This implies } \overrightarrow{FE} = \overrightarrow{FB'}$$

RAA Hypothesis

$$5) \text{ In that case, } B' = E$$

Contradiction

$$6) \text{ Hence, } \triangle ABC \cong \triangle DEF$$

27.

(pg 127) "Proposition 3.18"

"If in $\triangle ABC$ we have $\angle B \cong \angle C$,

then $AB \cong AC$ and $\triangle ABC$ is isoscles"

Proof: Hypothesis

Evidence

1) There is a unique point B' on ray DE such $DB' \cong AB$

Axiom C-1

$$2) \triangle ABC \cong \triangle DB'F$$

$$DB' \cong AB, \angle A \cong \angle D, AC \cong DF, \text{ Axiom C-6}$$

$$3) \angle DFB' \cong \angle C$$

Axiom C-5

$$4) \text{ This implies } \overrightarrow{FE} = \overrightarrow{FB'}$$

RAA Hypothesis

$$5) B' = E$$

Contradiction

6) Hence, $\triangle ACD \cong \triangle BCD$

Axiom C-4

7) Therefore, $\angle B \cong \angle C$

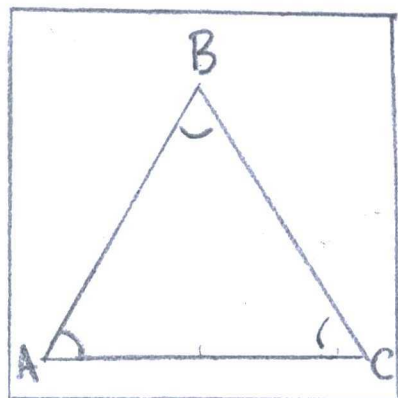
Proposition 3.10

8) $\triangle ABC$ is isosceles

(pg 31) I.5

28.

An Equiangular Triangle is Equilateral.



"Equiangular Triangle"

Proof: Hypothesis

Evidence

1) $\angle B \cong \angle C, \angle A \cong \angle B,$

Conjecture to Equiangular

$\angle A \cong \angle C$

2) $AB \cong AC$

Converse to Proposition 3.10

3) $\triangle ABC \cong \triangle ACB$

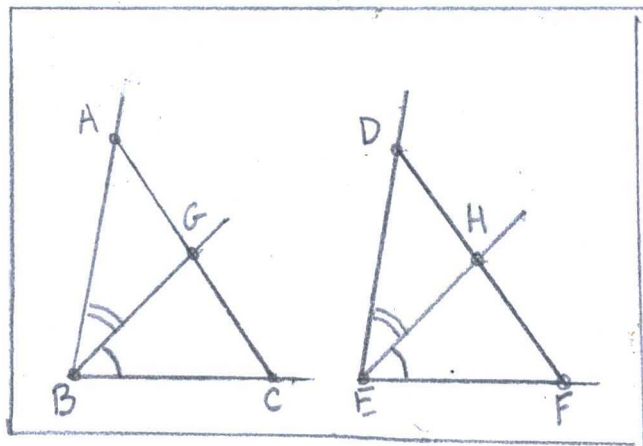
Implication by RAA

4) $AB \cong AC, BC \cong AC,$
 $AB \cong BC$

Equilateral

29.

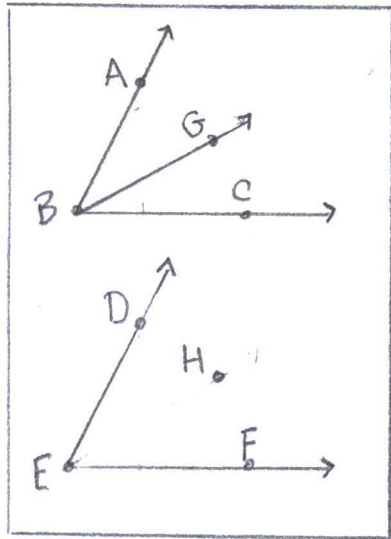
(pg 127) "Proposition 3.20
(Angle Subtraction)"



"Given $\overrightarrow{BG}$ between BA and BC , $\overrightarrow{EH}$ between ED , EF ,
 $\angle CBG \cong \angle FEH$, and $\angle ABC \cong \angle DEF$
Then $\angle GBA \cong \angle HED$."

Proof:	Hypothesis	Evidence
	1) $\angle ABC \cong \angle DEF$	Converse Proposition 3.19
	2) $\angle GBA \cong \angle HED$	Axiom C-4

30.



$\angle ABC \cong \angle DEF$ and $\overrightarrow{BG}$ between $\overrightarrow{BA}$ and $\overrightarrow{BC}$
implies $\overrightarrow{EH}$ between $\overrightarrow{ED}$ and $\overrightarrow{EF}$:

Proof	Hypothesis	Evidence
	1) A point H is between ED and EF	Proposition 3.7
	2) $AG \cong DH$	Proposition 3.12
	3) $AB \cong DE$	Axiom C-1
	4) $\angle CBG \cong \angle FEH$	Contrapositive to Proposition 3.19
	5) $\angle ABC \cong \angle DEF$	False contrapositive
	6) $\angle ABG \cong \angle DEH$	Proposition 3.20

31.

(pg 128) "Proposition 3.21 (Ordering of Angles)"

a) Exactly one of the following three conditions (trichotomy): $\angle P < \angle Q$, $\angle P \cong \angle Q$, or $\angle Q < \angle P$

b) If $\angle P < \angle Q$ and $\angle Q \cong \angle R$, then $\angle P < \angle R$

Case 5:

1) A ray $\vec{p}$ is
between $\angle P$, so
 $\angle P \cong \angle R$

(pg 126) "Definition -
Angle
Inequality"

Case 6:

1) A ray $\vec{r}$ is
between $\angle R$,
so $\angle P \cong \angle r$

(pg 126) "Definition -
Angle
Inequality"

32.

(pg 128) "Proposition 3.22 (SSS criterion for congruence)"

"If $AB \cong DE$, $BC \cong EF$, and $AC \cong DF$, then $\triangle ABC \cong \triangle DEF$ "

Proof: Hypothesis

Evidence

1) Point F is on DE
so $\triangle ABC \cong \triangle DEF$

Corollary to
Axiom C-6

2) $\angle A \cong \angle D$ and
 $\angle C \cong \angle F$

"Proposition 3.17
(ASA Criterion
for congruence)"

33.

(From exercise)

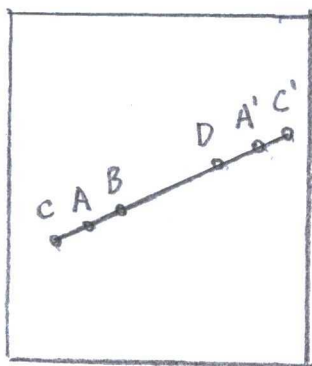
"If $AB < CD$, then $\angle A < \angle C$ "

Proof: Hypothesis

Evidence.

1) Points A, B, C, D,
A', and C' are on
a line

Proposition 3.3



2) Point $B = D$

3) $AB \cong BA'$

4) $CB \cong BC'$

5) $AA' < CA'$

6) $CA' < CC'$

7) $AA' \equiv 2AB$ and

$CC' \equiv 2BC$

8) $2AB < 2BC$

9) But point B is not contradiction

Also $2AB < 2CD$ |

Conjecture to
an RAA

Axiom C-1

Axiom C-1

Proposition 3.21

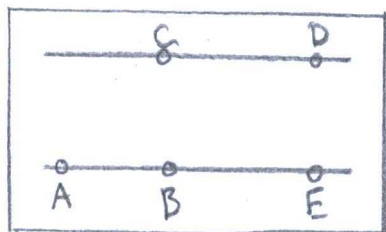
Proposition 3.21

(pg 65) "Logic Rule-9

$P \rightarrow Q \rightarrow R$

34.

a) (pg 16) "Euclid's Postulate II"



"Euclid's
Postulate II"

"For every segment AB and for every segment CD there exists a unique point E on line AB such that B is between A and E and segment CD is congruent to segment BE ."

Proof:

Hypothesis	Evidence
1) $CD \cong BE$	Axiom C-1
2) Another point A is on BE For $A * B * E$.	Axiom B-2

b) (from exercise)

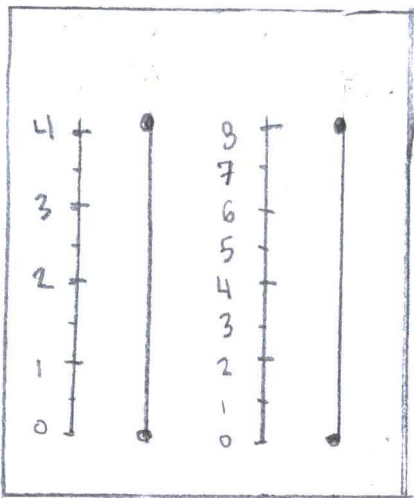
"The center of a circle is unique and its radius is unique up to congruence"

Proof:

Hypothesis	Evidence
1) Points O and O' exist with radius OA and OA'	Euclid's Postulate III
2) Circles are unique unless $O=O'$ and $OA=OA'$	$(\text{Circle } O) \cap (\text{Circle } O')$

35.

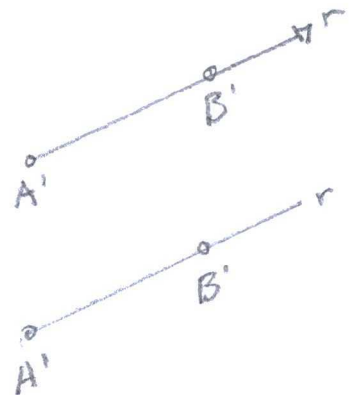
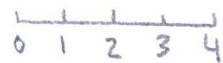
(From exercise)



"Two segments of the same length have the same length in this perverse way of measuring."

Informal proofs describing Axioms C-1/2/3/4/5

Axiom C-1:

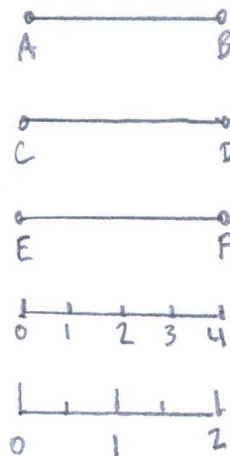


c) If $\angle P > \angle Q$ and $\angle Q \cong \angle R$,
then $\angle P > \angle R$

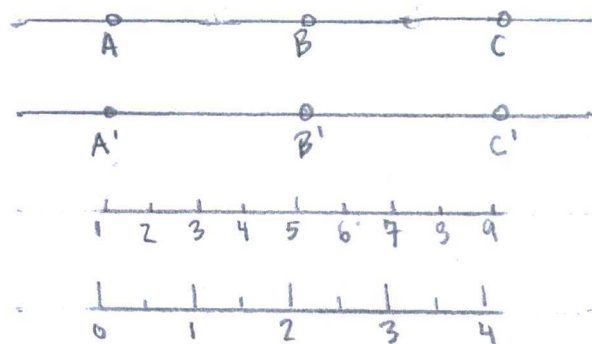
d) If $\angle P < \angle Q$ and $\angle Q < \angle R$, then
 $\angle P < \angle R$

Proof:	Hypothesis	Evidence
Case 1:	1) An angle $\angle R$ where $\angle R \cong \angle P$ 2) and $\angle R \cong \angle Q$ 2) $\angle P \cong \angle Q$	Axiom C-4
Case 2:	1) An angle $\angle R$ where $\angle R \cong \angle P$ and $\angle R < \angle Q$ 2) $\angle P < \angle Q$	Axiom C-4
Case 3:	1) An angle $\angle R$ where $\angle R \cong \angle P$ and $\angle R > \angle Q$ 2) $\angle P > \angle Q$	Axiom C-4
Case 4:	1) A ray $\vec{r}$ is in $\angle R$, so $\angle P \cong \angle r$	(Pg 128) "Definition- Angle Inequalities."

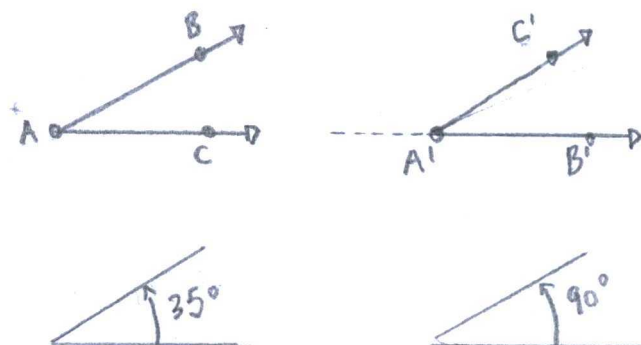
Axiom C-2:



Axiom C-3:

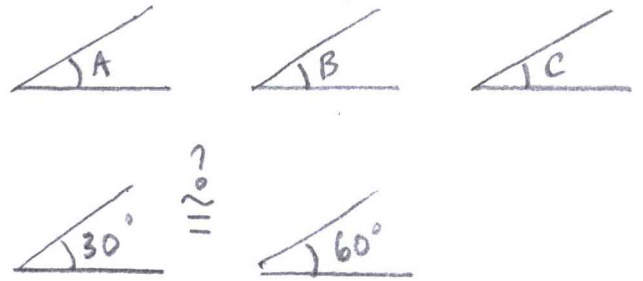


Axiom C-4:

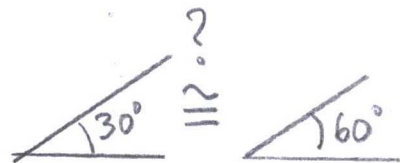
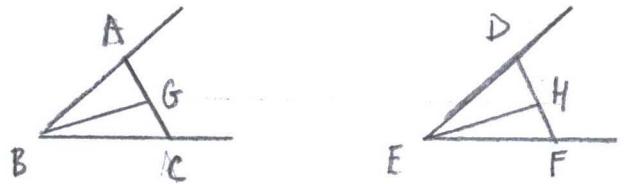


Note: Axiom C-4 may be false to mathematicians with "perverse" angles. The total degrees is constant around a circle from radians relationship to degrees. Base-60 describes π in a series.

Axiom C-5:



Proposition 3.19:



Axiom C-6 Fails with all triangles
because prerequisites, including
sum total angles. and

Circle-circle continuity principle
fails because 360° rotation.

Segment-circle principle cannot follow
the "perverse" measurement
because absolute degrees inside
a circle.